T20 retained-record review

2026-10-20 · Crypto PR #2318 · review addendum

This addendum qualifies the historical n = 73, K = 600 comparison. Original protocol, measurements and reports remain unchanged. The native analyzer reproduces the retained JSON, and the three IC answers independently replay.

Timing provenance correction

The historical prose says rho draw 1 was kept from before IC. The retained rho_01.time instead matches its second isolation receipt:

Receipt	UTC start	Wrapper wall seconds
Earlier	Oct 19, 19:08:28	163.95028786
Retained	Oct 20, 03:17:31	105.25283527

Draws **1–12** have duplicate receipts. All twenty retained timings match their final receipts and follow the completed IC runs. None brackets IC. The 41 original isolation receipts remain present, with exit zero and contended=false under the guest-local monitor.

Interpretation

The 2.027 old/new wall ratio reproduces from the retained data. Its bootstrap interval is conditional on those runs and their columns. Two same-arm runs provide one observed A/A difference; they do not bound run-to-run noise. There is only one old-arm run.

Recorded probes, attempts, relations and answers agree across the three arms. The three-operation-per-probe proxy is reproducible. Complete charged-cost equality remains **unmeasured**: IC group additions and operation-kind counts were not emitted. The live leaderboard separates the proxy ratio from the null complete-cost ratio. A zero lower bound at weight zero is not a measurement of zero total work or evidence for the cross-method ordering.

Analyzer validation

The original analyzer returned success for six deliberately invalid inputs: unverified IC, altered N2 relation, altered N2 target, unverified rho, altered rho target and a missing rho draw. It still emitted kill_triggered=false. The repair requires verified fixed-identity inputs, all twenty rho draws, unique rank columns and complete finite timing/counter fields before emitting analysis. The six controls now fail; the retained dataset's JSON is unchanged and four native regression tests pass.

This review does not authorize another timing run. Parent #2156, native producer validation and current-head CI remain merge gates. See T20_REVIEW_20261020.md for source paths and the precise evidence boundary.